

Exp. 5. Band Structure Simulation for Graphene and CNTs using NanoHub

Case Study 1: Simulate the band-structure of Si circular nanowire

Homework Bandstructure Lab x Band Structure Lab

1 Device Type → 2 Physics → 3 K-Space → 4 Numerics → 5 Simulate

Geometry: Nanowire (1D-periodic)

Calculation For: Electrons

Material: Si

Alloy Model: Virtual Crystal Approximation

Molar fraction: 0.5

Job Type: Calculate the wire band structure

Device cross section: Circle

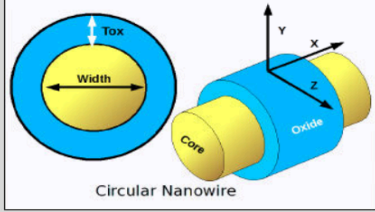
Diameter: 2.1nm Choose the geometry of the cross section

Oxide Thickness(T_{ox}): 2.2nm

Crystal Orientation

Transport direction (X): (100)

Confinement direction (Z): [010]



Circular Nanowire

Physics >

Storage (manage) 0% of 10.000004GB

900 × 800

ABACUS - Assembly of Basic Applications
for Coordinated Understanding of
Semiconductors

Terminate Keep for later

Assembly of
Basic Applications for
Coordinated Understanding of Semiconductors

Choose Applications and Tools in this Dropdown Menu or from Images

Bandstructure Lab - Bulk, quantum wells, nanowires

Homework Bandstructure Lab x Band Structure Lab

1 Device Type → 2 Physics → 3 K-Space → 4 Numerics → 5 Simulate

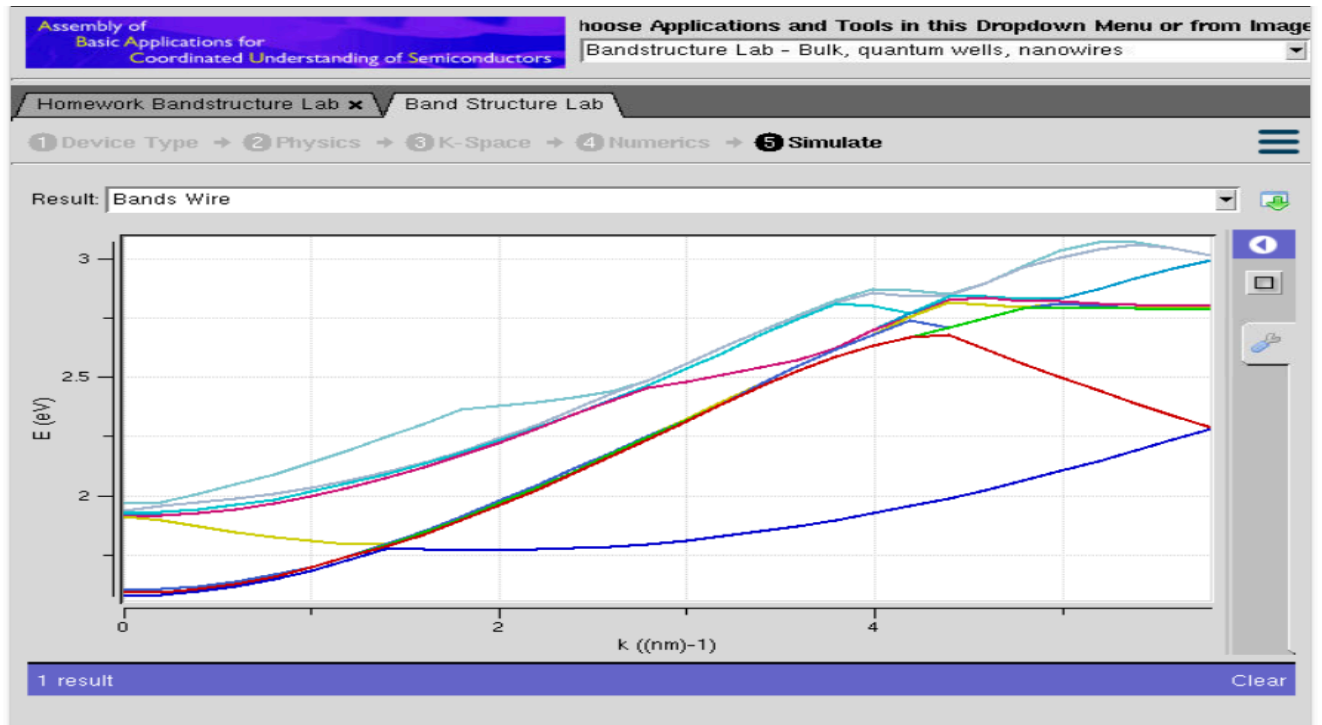
Electronic Structure Strain

Tight Binding Model: sp3d5s*

Spin-Orbit Coupling: ☒ no

Dangling Bond Energy: 30eV

Explore: Conduction bands



Case Study 1: Simulate the band-structure of InAs circular nanowire

1 Device Type → 2 Physics → 3 K-Space → 4 Numerics → 5 Simulate

Geometry: Nanowire (1D-periodic)

Calculation For: Electrons

Material: InAs

Alloy Model: Virtual Crystal Approximation

Molar fraction: 0.5

Oxide Thickness(T_{ox}): 2.2nm

Crystal Orientation

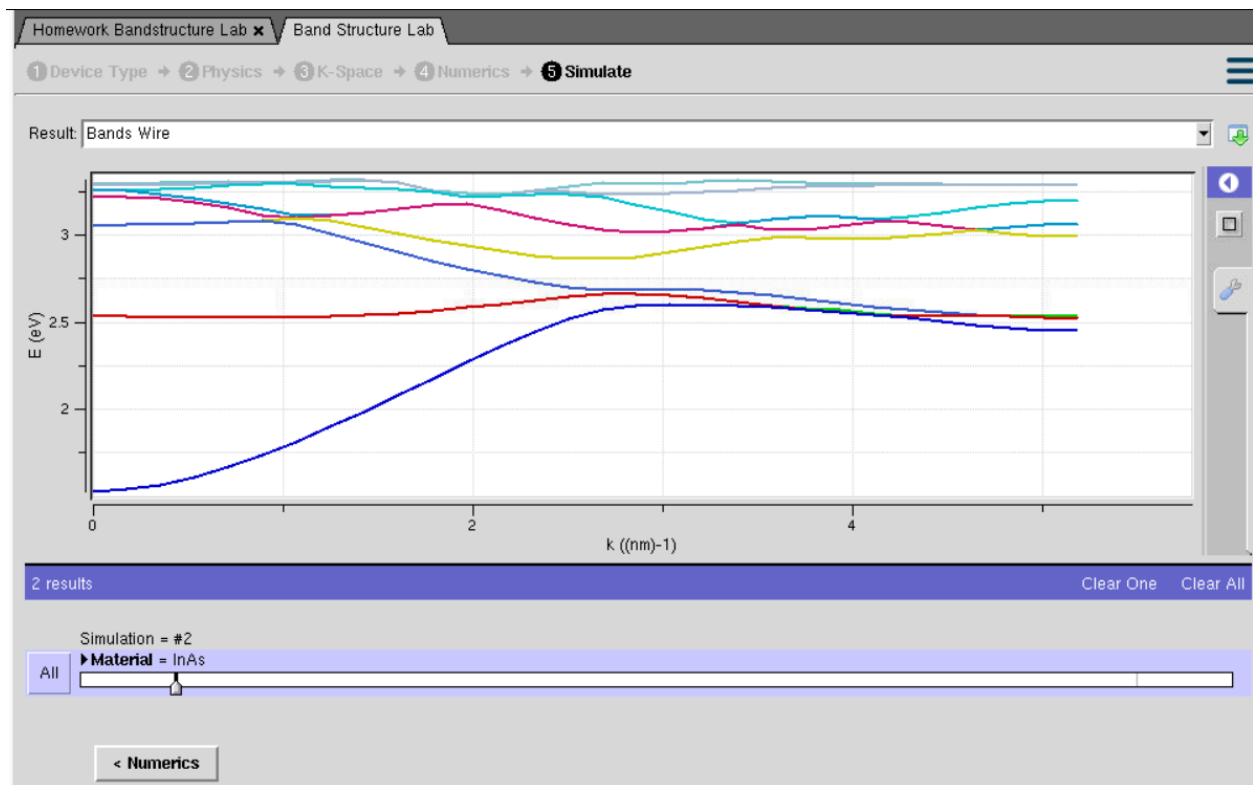
Transport direction (X): (100)

Confinement direction (Z): [010]

Circular Nanowire

Physics >

Storage (manage) 0% of 10.000004GB 954 × 666



Case Study 2: Simulate the band-structure of AlGaAs circular nanowire

1 Device Type → 2 Physics → 3 K-Space → 4 Numerics → 5 Simulate

Geometry: Nanowire (1D-periodic)

Calculation For: Electrons

Material: AlGaAs

Alloy Model: Virtual Crystal Approximation

Molar fraction: 0.5

Oxide Thickness(T_{ox}): 2.2nm

Crystal Orientation

Transport direction (X): (100)

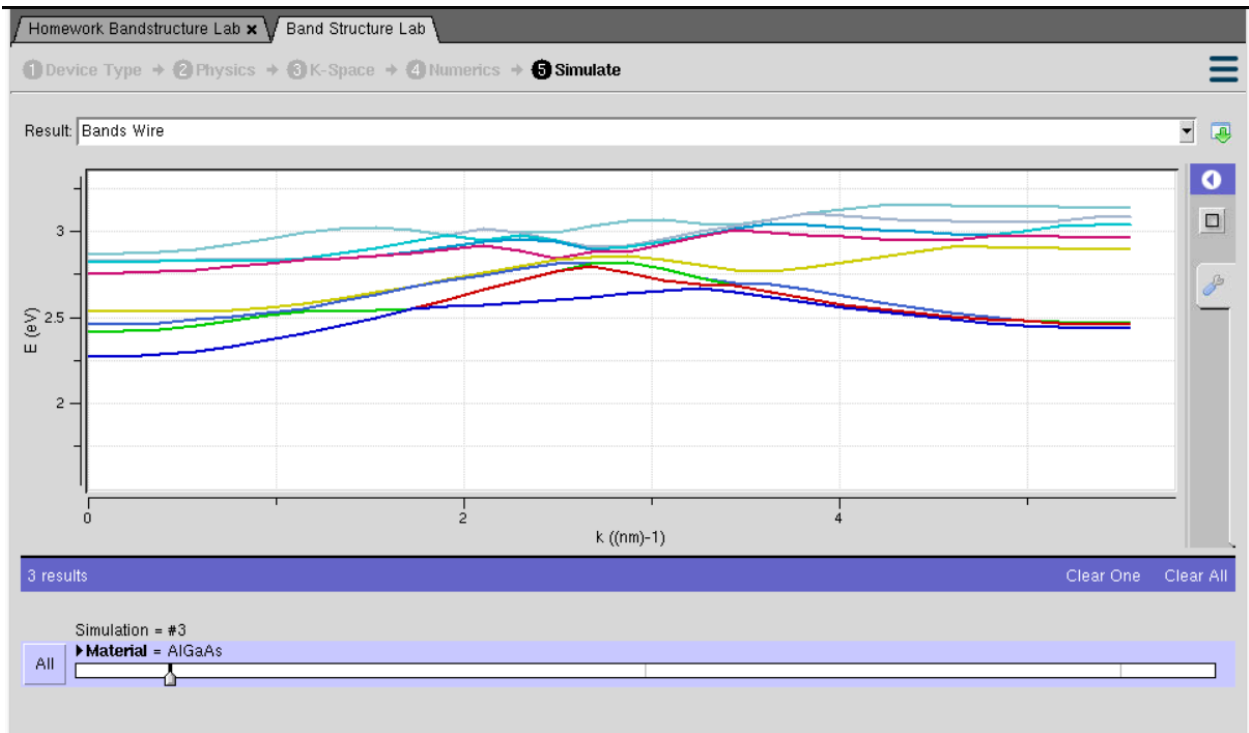
Confinement direction (Z): [010]

Circular Nanowire

Physics >

Storage (manage) 0% of 10.000004GB

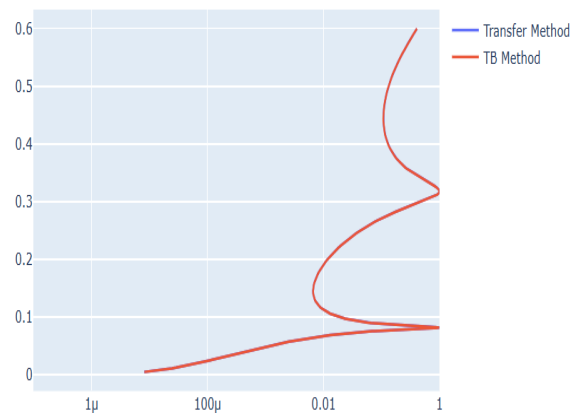
954 × 666



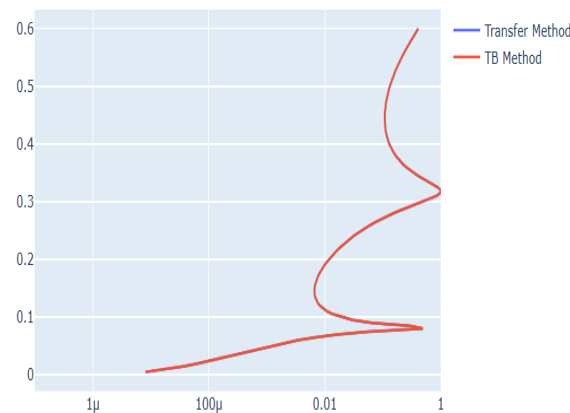
Exp. 6. Demonstration of Quantum Tunneling

Case study 1 : Analysis the transmission co-efficient Vs Energy characteristics on refined grid

Transmission Coefficient Vs Energy on refined grid

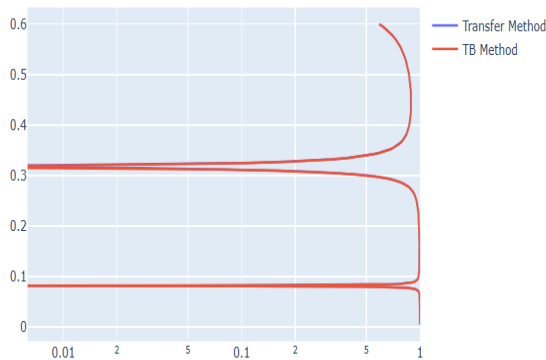


Transmission Coefficient Vs Energy on Homogeneous Grid

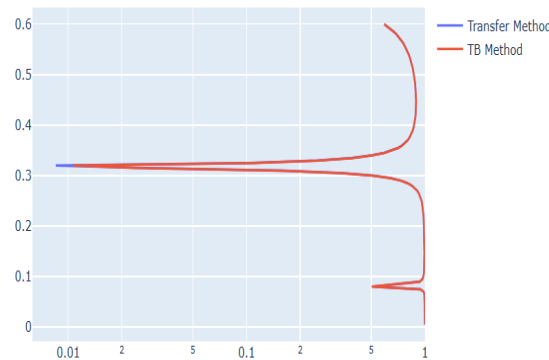


Case study 2 : Analysis the Reflection co-efficient Vs Energy on Homogeneous Grid

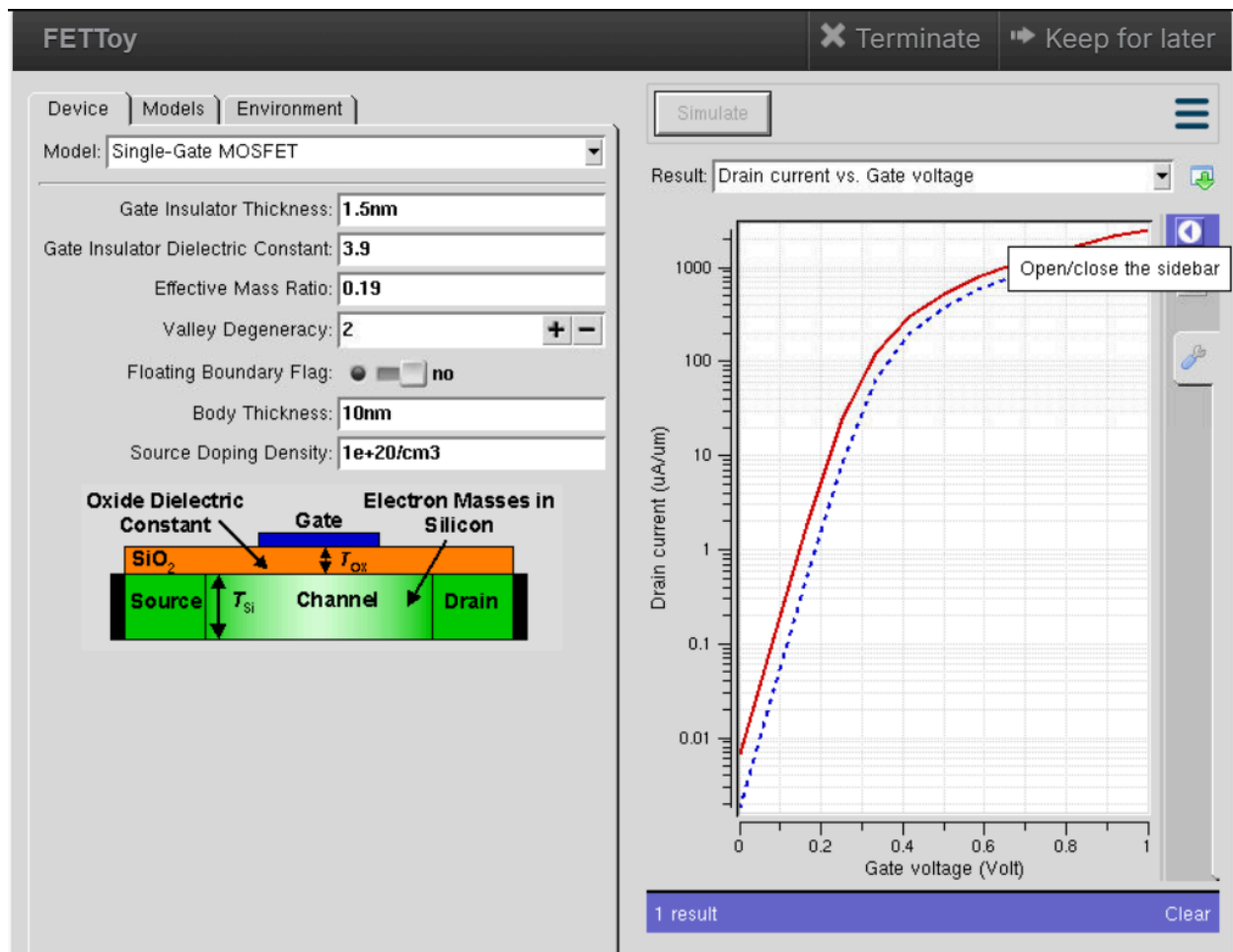
Reflection Coefficient Vs Energy on refined grid



Reflection Coefficient Vs Energy on Homogeneous Grid



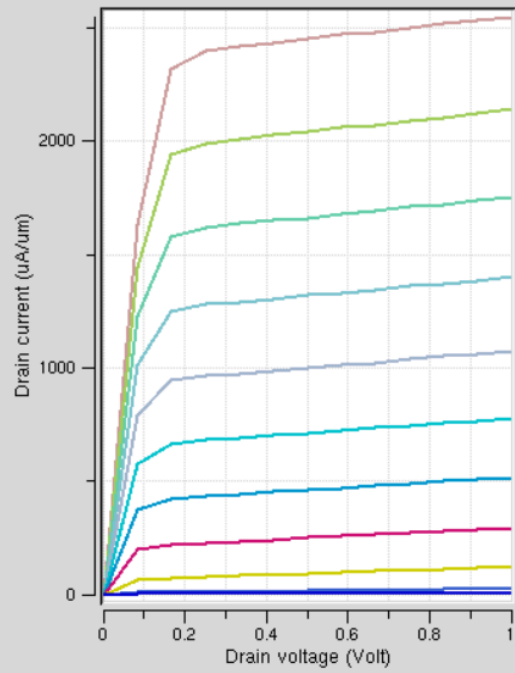
Exp. 7. MOSFET Scaling Analysis and Subthreshold Swing Calculation using NanoHub



Device Models Environment

Simulate

Result: Drain current vs. Drain voltage



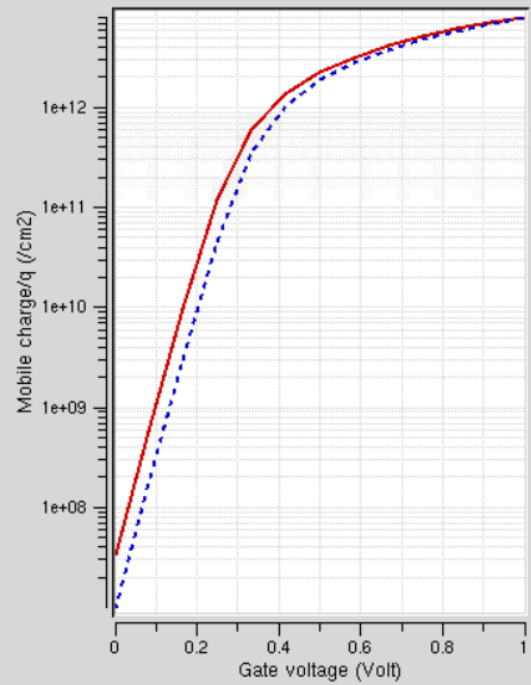
1 result

Clear

Device Models Environment

Simulate

Result: Mobile charge/q vs. Gate voltage



1 result

Clear